

# Bank System API - Full Project Documentation

## 1. Project Overview

This project is a secure banking system backend built with Flask, JWT authentication, and SQLite database. It supports user registration, login, deposits, withdrawals, transfers, and admin analytics.

## 2. Tech Stack

Flask, SQLite, JWT (PyJWT), Python, Render deployment, Swagger UI, Werkzeug security.

## 3. System Features

- User authentication (register/login)
- JWT-based security
- Deposit & withdraw money
- Transfer between users
- Transaction history
- Admin dashboard endpoints

## 4. API Endpoints

/register - Create user  
/login - Authenticate user  
/deposit - Add money  
/withdraw - Remove money  
/transfer - Send money  
/transactions - View history  
/admin/users - Admin user list  
/admin/transactions - Admin transaction logs

## 5. Architecture

Flask backend with layered architecture:  
Routes -> Services -> Database  
JWT middleware protects secured endpoints.

## 6. Deployment

Deployed on Render using Gunicorn WSGI server. Environment variables used for SECRET\_KEY.

## 7. Security

Password hashing using Werkzeug, JWT authentication, admin role protection middleware.

## 8. System Architecture Diagram

